

3. Metals have a variety of applications in industry, including making the wings and wiring for aeroplanes.

(a) (i) Describe metallic bonding. [2]

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(ii) Explain the property of metals which allows them to be used to make sheets for aeroplane wings. [2]

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(iii) Explain why the metals used in wires are good conductors of electricity. [2]

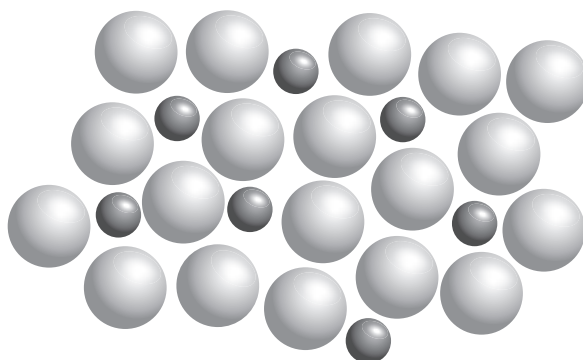
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- (b) Wings of modern jet fighter aircraft are made from alloy X which is composed of titanium, aluminium and vanadium. Wings on older aircraft were made from steel.

(i) Alloys consist of more than one type of atom, as shown in the diagram below.



Explain why this increases the strength. [2]

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- (ii) Some properties of these metals and alloys are given in the table below.

Metal	Density (kg/m ³)	Stiffness (GPa)	Tensile strength ($\times 10^7$ Pa)
titanium	4 500	110	55
aluminium	2 700	69	10
vanadium	5 700	138	63
alloy X	4 400	110	100
steel	7 800	210	40

- I Compare the use of alloy X with steel for making aeroplane wings. [2]

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- II The mass of a wing made from alloy X is 2 500 kg. The alloy is 90 % titanium and 6 % aluminium.
Use the equation:

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

- to calculate the volume of vanadium needed for each wing. [4]

Volume = m³